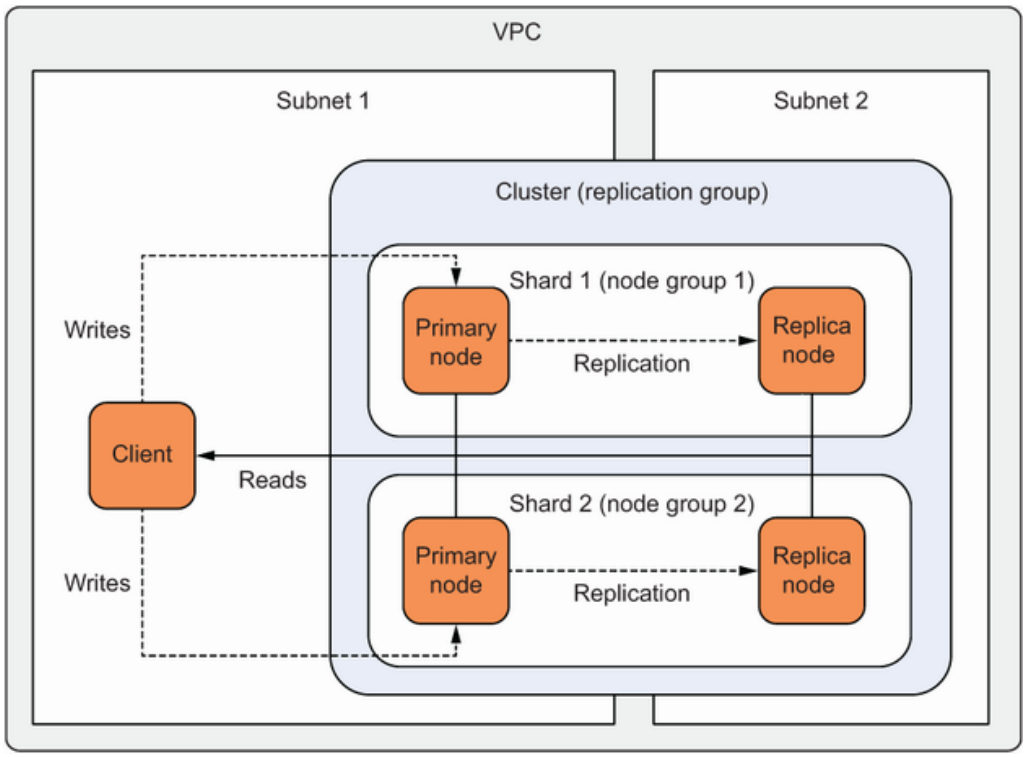


Figure 11.7 Redis deployment option: Cluster with cluster mode enabled

Use a Redis cluster with cluster mode enabled when you need data replication and your data is too large to fit into the memory of a single node. Imagine that your cached data is 22 GB in size. Each cache node has a capacity of 4 GB of memory. Therefore, you will need six shards to get a total capacity of 24 GB of memory. ElastiCache provides up to 437 GB of memory per node, which totals to a maximum cluster capacity of 6.5 TB (15 * 437 GB).

Additional benefits of enabling cluster mode

With cluster mode enabled, failover speed is much faster, because no DNS is involved. Clients are provided a single configuration endpoint to discover changes to the cluster topology, including newly elected primaries. With cluster mode disabled, AWS provides a single primary endpoint, and in the event of a failover, AWS does a DNS swap on that endpoint to one of the available replicas. It may take ~1–1.5 minutes before the application is able to reach the cluster after a failure, whereas with cluster mode enabled, the election takes less than 30 seconds.

On top of that, increasing the number of shards also increases the maximum read/write throughput. If you start with one shard and add a second shard, each shard now only has to deal with 50% of the requests (assuming an even distribution).